

Preeti Hegde

www.linkedin.com/in/preetivhegde

Würzburg, Germany • preetivhegde98@gmail.com

AI and Software Engineer with around 3 years of experience developing backend services and hands-on machine learning applications, Retrieval-Augmented Generation (RAG) systems, and AI Agents. Proven ability to integrate LLMs, semantic search, and data processing workflows into production systems. Combines strong ownership and an engineering-focused approach with a focus on scalability, maintainability, and quality.

Skills & Technologies

- **Languages:** Python, Java
- **Technologies & Frameworks:** PyTorch, LangChain, LangGraph, FastAPI, OpenAI API, SentenceTransformers, SQL, Vector Databases, Retrieval-Augmented Generation (RAG), Semantic Search, Prompt Engineering, Machine Learning, Predictive Modelling, Statistical Analysis, Spring Boot, REST APIs, Distributed Systems, Docker, AWS, Git, CI/CD
- **Expertise:** Generative AI, Large Language Models (LLMs), AI Application Development

Work Experience

AI and Machine Learning - Intern [Chi2 Data Analytics and Artificial Intelligence GmbH](#) **November 2025 - January 2026**
Würzburg, Germany (Remote)

- **Built Python- and analytical workflows for large research datasets**, using Pandas, NumPy, statistical modelling, and ggplot2 to identify trends, anomalies, outliers, and performance patterns.
- **Developed structured data validation and EDA pipelines across multiple datasets**, covering missing-value checks, outlier detection, consistency review, transformation logic, and statistical preparation.
- **Applied 5+ statistical and modelling techniques** including regression analysis, hypothesis testing, correlation analysis, significance testing, and anomaly detection.
- **Automated repetitive analysis tasks with reusable Python**, improving repeatability, reducing manual review effort, and supporting reproducible workflows.
- **Built visual reports and technical summaries** to explain trends, anomalies, and validation results to stakeholders.
- **Maintained clean documentation, traceable analysis steps, and reproducible outputs**, supporting data integrity, quality control, and reliable handover of analytical work.

Software Engineer [Altisource](#) **January 2021 - March 2023**
Bengaluru, India

- **Developed backend services for Equator, an enterprise mortgage workflow platform**, using Java, Spring Boot, Microservices, Flowable BPM, MySQL, REST APIs, SOAP services, JSON, XML, Kafka, and AWS.
- **Owned end-to-end delivery of bulk property onboarding automation using Spring Batch**, replacing manual UI-based entry with automated file processing, validation, transformation, and backend ingestion.
- **Built and maintained integrations with 12+ external vendor systems**, enabling workflow updates, task completions, and status changes to synchronize automatically between vendor platforms and Equator.
- **Developed 3+ microservices from scratch** and contributed to the migration of a 15-year legacy system toward a scalable microservices architecture.
- **Implemented automation services** for vendor communication, data synchronization, report retrieval, workflow triggers, retry handling, and operational follow-up reduction.
- **Integrated Stripe payments using REST APIs and webhooks**, automating transaction status updates, payment reconciliation, exception handling, and real-time backend workflow updates.
- **Built a reusable PDF generation engine using Apache FreeMarker templates**, converting application data into structured documents while preserving formatting, layout, and consistency.
- **Supported cloud deployment, monitoring, and troubleshooting across AWS IAM, ECS, and EKS**, working with Cloud, DevOps, Security, QA, Product, business teams, and external vendors.
- **Recognized as a Star Employee within 6 months** for strong ownership, support and collaboration.

Education

Master in Artificial Intelligence, Technische Hochschule Würzburg-Schweinfurt, Germany	2023 - 2025
Bachelor of Engineering: Computer Science and Engineering, Sri Siddhartha Institute of Technology, India	2016 - 2020

Projects & Extra

Conversational AI Assistant (Uttara AI)	March 2026 - May 2026
Developed a RAG-based conversational assistant using vector search, semantic retrieval, and LLM-powered response generation for portfolio data analysis.	

FileChatAI	January 2026 - March 2026
Built a document intelligence platform using FastAPI, LangChain, OpenAI, and Qdrant, enabling semantic search and question answering across PDF documents.	

Master's Thesis, Technische Hochschule Würzburg-Schweinfurt, Germany	November 2024 - May 2025
Developed a deep learning solution for image classification and anomaly detection in reusable container inspection using PyTorch and Docker.	

LLM Foundations: Vector Databases for Caching and Retrieval Augmented Generation (RAG), LinkedIn Learning	December 2025
Model Context Protocol (MCP) for Beginners by Microsoft, LinkedIn Learning	December 2025

Languages

- English: Full Professional Proficiency C1
- German: Intermediate Proficiency (B1) actively improving